🧭 M003 Guide — Counting (x, y, z) · Intermediate

Topic: Before 3D · **Course:** 3D Coordinates · **Level:** Intermediate · **Time:** about 25 minutes

Pixel art used **two** numbers. A 3D build needs **three**. They always come in the same order:

1st = X	2nd = Y	3rd = Z
right → left	down → up	back → forward
starts at 0	starts at 0	starts at 0

All three rulers meet on **one block** — your home spot. That block is **(0, 0, 0)**, and every count starts there.

> 🔴 Stand on your **home spot** (red block) every run. Move your feet, move your build.

1 · Build Your Grid 🎨

Make a striped ruler for each axis, so you can **see** the numbers.

The floor starts at the corner **(0, -1, 0)** — one below your feet — and reaches 15 blocks left and 15 blocks forward. Fill in the missing corners.

```
def on_run():
    blocks.fill(WHITE_CONCRETE,
        pos(0, -1, 0),
        pos(__, -1, __),
        FillOperation.REPLACE)
    # yellow line 5 to the left
    blocks.fill(YELLOW_CONCRETE,
        pos(5, -1, 0),
        pos(5, -1, 15),
        FillOperation.REPLACE)
    # yellow line 5 forward
    blocks.fill(YELLOW_CONCRETE,
        pos(0, -1, __),
        pos(15, -1, __),
        FillOperation.REPLACE)
player.on_chat("grid", on_run)
```

Why is the floor at y = -1 and not y = 0?

Now place the three rulers by hand — **yellow, black, yellow, black** all the way. All three start on the corner block:

- **X ruler** — one block at a time, **to your left**.
- **Z ruler** — one block at a time, **forward**.
- **Y ruler** — a pillar straight **up**.

2 · Predict

Read each set of blocks. Circle the direction, then say which number changed.

```
place red block at (1, 0, 0)
place red block at (2, 0, 0)
place red block at (3, 0, 0)
```

Circle one: left · up · forward

Which number changed? Circle one: x · y · z

```
place red block at (0, 0, 1)
place red block at (0, 0, 2)
place red block at (0, 0, 3)
```

Circle one: left · up · forward

Which number changed? Circle one: x · y · z

```
place red block at (7, 1, 2)
place red block at (7, 2, 2)
```

Circle one: left · up · forward

Where is the corner block? Write its coordinate: (_ , _ , ___)

3 · Spot the Bug 🐛

Each one is broken. Write the fixed code, then explain.

Bug A — This should be a pillar 3 blocks **up** the y ruler. Right now the same block is placed three times, because the **2nd number** never changes.

```
place red block at (0, 1, 0)
place red block at (0, 1, 0)
place red block at (0, 1, 0)
```

Write the fixed code:

Why was it wrong? Why does your fix work? (1–2 sentences.)

Bug B — Your friend says the corner block is **(1, 1, 1)**, so they start counting there. Their whole build sits one step left, one block up, and one step forward from where it should be.

```
place red block at (1, 1, 1)   # they think this is the corner
```

Write the corner's real coordinate:

Which number is floating their build off the ground? Circle one: x · y · z

4 · Put Them Together 🧭

Read **(5, 2, 3)** in three steps:

Step	Number	Ruler	Do this
1	x = 5	right → left	go 5 to your left
2	y = 2	down → up	go 2 up (the ground is 0)
3	z = 3	back → forward	go 3 forward

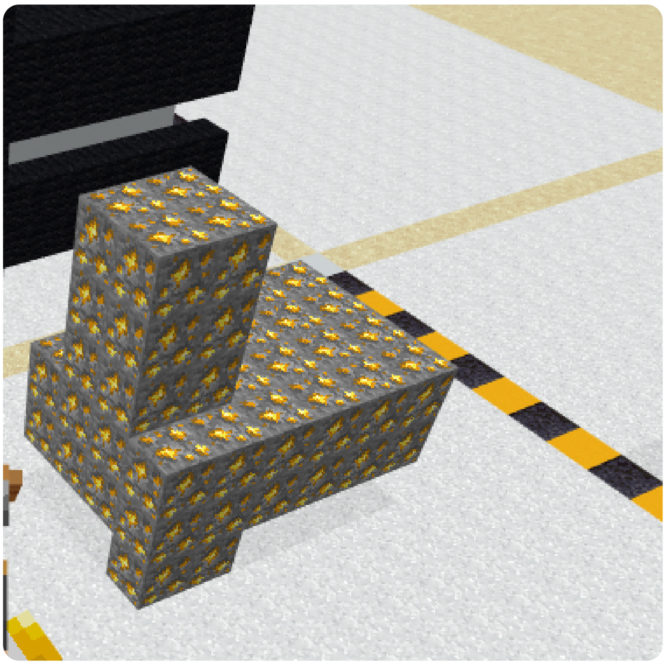
Which coordinate is on the ground? Circle one: (2, 1, 2) · (2, 0, 2)

A block sits 4 to your left and 6 forward, on the ground. Write it: (_ , _ , ____)

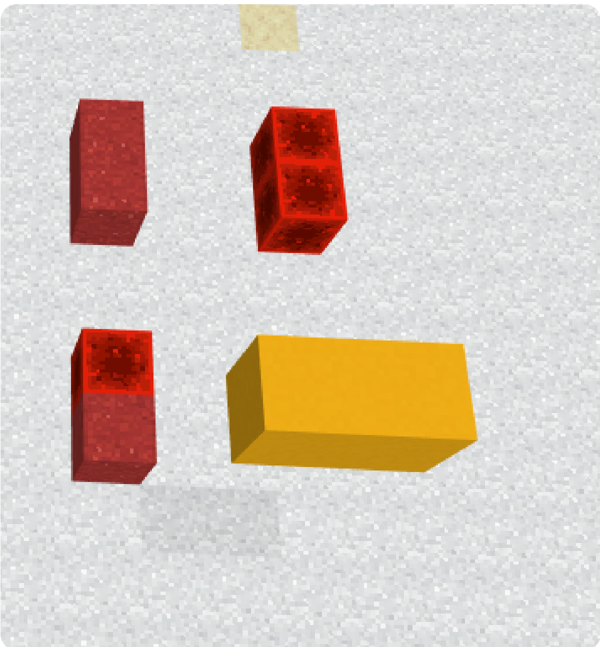
5 · Show Your Work 📸 🎥

Warm-up: stand on your home spot and place one block at **(3, 0, 3)**. Check it against your rulers.

Open the world. Part of the shape is **already built** for you:



The rest of the blocks are **missing**. Here they are, spread apart so you can see each one:



Finish the shape. Read each missing block off the model and count from the corner — x, then y, then z.

Write the coordinates you counted, then the code you ran.

> 🧱 The corner block is **(0, 0, 0)**. Use the same blocks you see in the world.

Record **one video** — one take, no stopping (a phone is fine). Show these in order:

1 · Your code. Scroll through it. Say what each part does.

2 · Your build. Point the camera. Count one block out loud — x, then y, then z.

Fill the blanks:

> Today I built _____.
>
> I built it using this code: _____.
>
> In this code I used _____.
>
> The hardest part was _____.
>
> That part was hard because _____.
>
> The most fun part was _____.
>
> Something new I learned was _____.

Write your lines here, then say them in your video.

Submit ✅

Send this worksheet + your video to teacher on KakaoTalk.